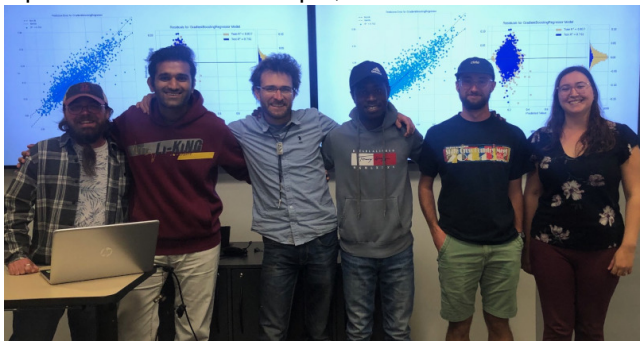
Same/Other/All K-Fold Cross-Validation for Estimating Similarity of Patterns in Data Subsets, Statistical Analysis and Data Mining (2026)

Toby Dylan Hocking — toby.dylan.hocking@usherbrooke.ca
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Introduction: how to evaluate machine learning algorithms?

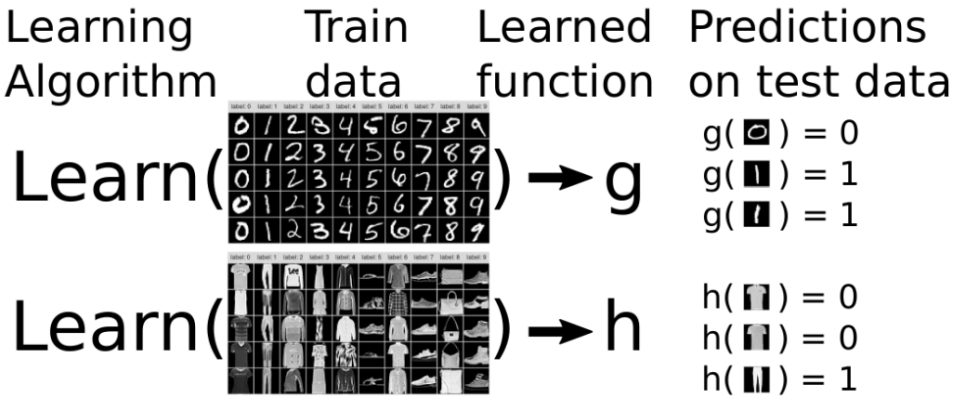
Proposed new algorithm, Same/Other/All K-fold cross-validation (SOAK)

Results on 20 data sets

Conclusions

Machine learning powers modern AI systems

Figure from chapter by Hocking TD, *Introduction to machine learning and neural networks* for book *Land Carbon Cycle Modeling: Matrix Approach, Data Assimilation, and Ecological Forecasting* edited by Luo Y (Taylor and Francis, 2022).



Learn is a learning algorithm, which outputs g and h.

Q: what happens if you do $g($  $)$, or $h($  $)$?

Train on one subset and accurately predict on another?

- Predicting medical diagnosis (Lindly *et al.*), train on one year of surveys, test on another. (**different time periods**)
- Algorithmic fairness: predicting mortality (Laribi *et al.*), train on men, test on women. (**different sex, age, etc.**)
- Predicting trees/burn/fish/carbon emissions (Aslam, Shenkin, Thibault, Bodine, *et al.*), train on one geographic area, test on another. (**different geographic regions**)
- Cybersecurity: predicting vulnerabilities from source code (Barr *et al.*), or attacks from network monitoring data, in **different software/sites/clients/etc.**
- These are questions about **generalization**: how accurate is the learned function on a new/test data subset which is **qualitatively different** in some respect?
- In theory, "should be accurate if test data are similar to train data."
- In practice, how to check for a particular data set and learning algorithm? Proposed method: Same/Other/All K-fold cross-validation (SOAK).

Introduction: how to evaluate machine learning algorithms?

Proposed new algorithm, Same/Other/All K-fold cross-validation (SOAK)

Results on 20 data sets

Conclusions

Research paper and software

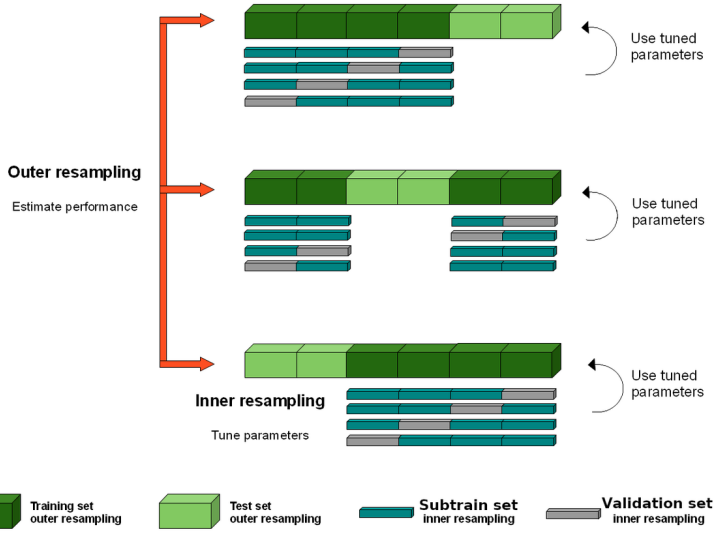
Toby Dylan Hocking, Gabrielle Thibault, Cameron Scott Bodine, Paul Nelson Arellano, Alexander F Shenkin, and Olivia Lindly. SOAK: Same/Other/All K-fold cross-validation for estimating similarity of patterns in data subsets. Statistical Analysis and Data Mining, 19(1), 2026.

Corresponding R package: mlr3resampling.
`https://mlr3resampling.netlify.app`

Standard K-fold cross-validation is used throughout machine learning

- For evaluation (outer), we split full data set into
  - **Training** or **train** set, used as input to learning algorithm
  - **Test** set, used to evaluate predictions of learned model
- For training (inner), we split training set into
  - **Subtrain*** set, used to compute gradients (etc.)
  - **Validation** set, used to avoid overfitting (tune best number of epochs, etc.)

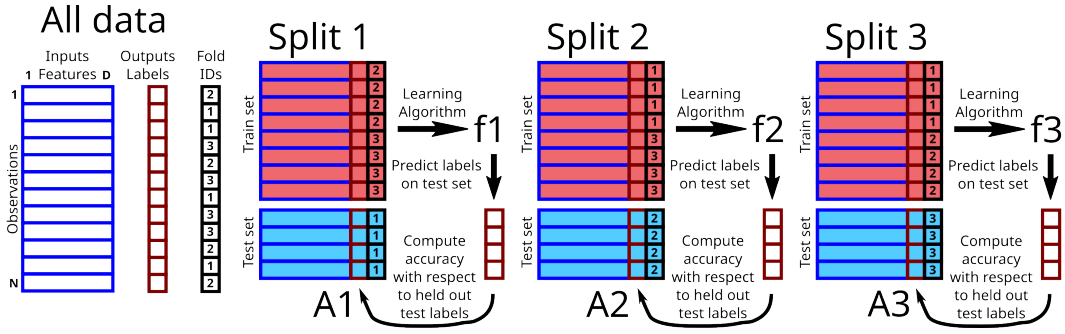
*Sometimes people say the inner split creates validation and another set which is also called "training" (confusing — I recommend calling it "subtrain" to clarify).



https://mlr.mlr-org.com/articles/tutorial/nested_resampling.html

Details of standard K-fold cross-validation

- $K = 3$ folds shown below: three different models trained, three different prediction/test accuracy rates computed.
- Important: several train/test splits allow computation of P-values, to test for statistically significant differences between algorithms (not possible with only one split).



Hocking TD *Intro. to machine learning and neural networks* (2022).

Example data with subsets: predicting childhood autism

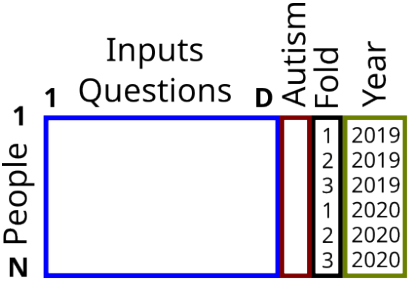
- Collaboration with Lindly *et al.*
- Downloaded National Survey of Children's Health (NSCH) data, years 2019 and 2020, from `http://www2.census.gov/programs-surveys/nsch`
- One row per person ($N = 46,010$ rows), one column per survey question ($D = 366$ columns).
- Output/label column is diagnosis with Autism (binary classification, yes or no), can we predict it using the 365 inputs/features?
- 18,202 rows for 2019; 27,808 rows for 2020.

Proposed SOAK algorithm is a generalization of standard K-fold cross-validation, that can be used to answer two related questions:

- Can we train on one year, and accurately predict on another?
- Can we get a more accurate model by combining data from different years?

Proposed SOAK algorithm (Autism data example)

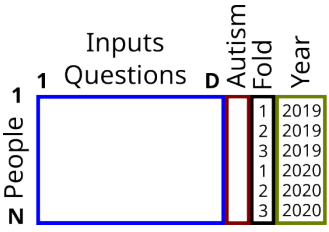
- Example: childhood autism prediction data set.
- Subsets of interest are years, which can be represented by adding a new column to the data table.

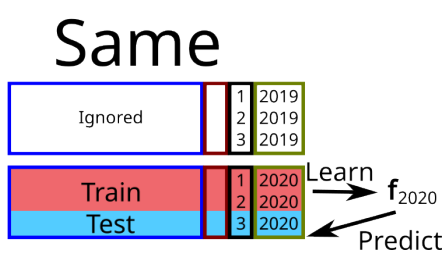


Proposed: Fold and Year are two different columns.
(In random Fold assignment, stratify by Year,
so there are equal numbers of rows of each fold in each year)

Proposed SOAK algorithm (Autism data example)

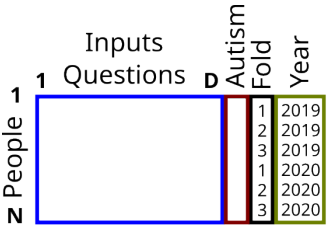
- Train subset same as test (=regular K-fold CV on 2020).
- Do we get a more or less accurate model than this baseline Same model? (if we train on Other/All years)

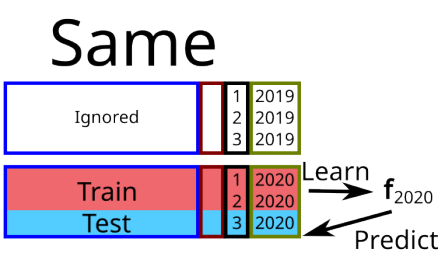


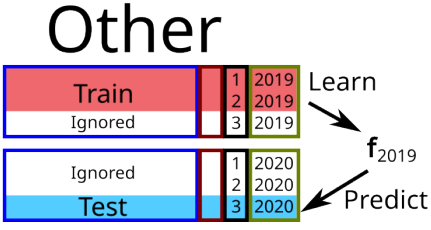


Proposed SOAK algorithm (Autism data example)

- Test subset fixed (2020), train on other subset/year (2019).
- Can we train on one year, and accurately predict on another? Compare Same/Other test error.



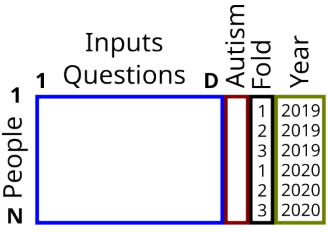


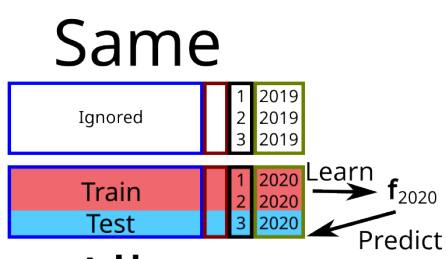


— Is Other as accurate as Same?

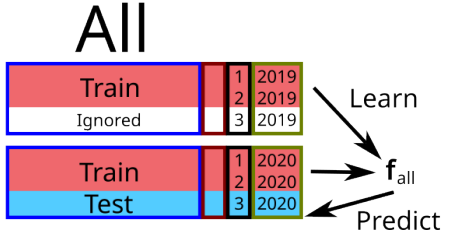
Proposed SOAK algorithm (Autism data example)

- Train set includes data from both subsets/years (2019, 2020).
- Can we get a more accurate model by combining data from different years? Compare Same/All test error.



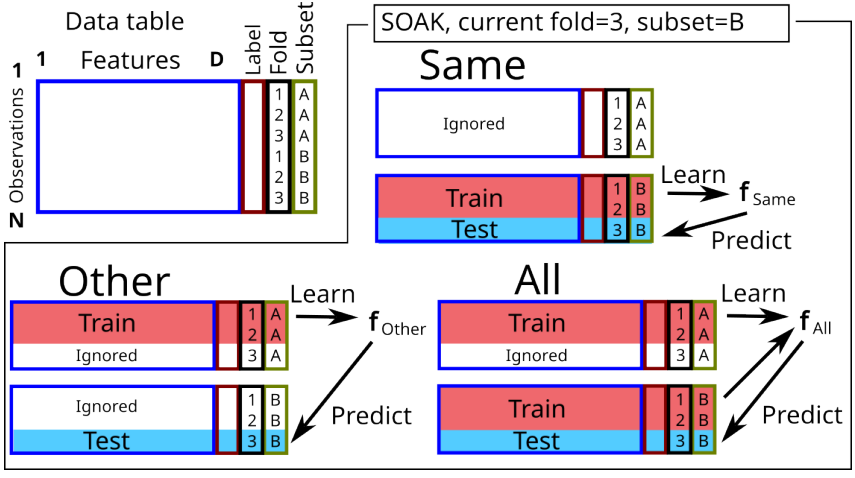


— Is All as accurate as Same?



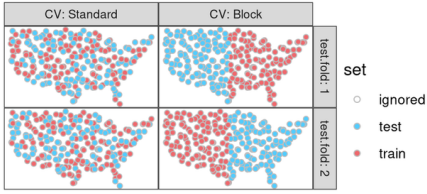
Proposed SOAK algorithm (generic data)

- Key new idea is **subset** column in data table.
- Example: $K = 3$ folds, two subsets (A/B).
- Compute test error for each fold (1/2/3) and subset (A/B).



Proposed SOAK algorithm (geographical data example)

Previous: standard and blocked train/splits



Significant difference between Standard and block CV? Wenger and Olden (2012), Roberts et al. (2017), Valavi et al. (2019), Ploton et al. (2020), Schratz et al. (2024).

— Is predicting more difficult when folds are assigned spatially rather than randomly?

— Previous papers do not compute P-values, although un-paired T-test could be used (less powerful).

— Block variable (West/East) used as fold.

— Number of folds must be equal to number of blocks (K=2 folds here for two subsets, West and East).

Proposed: SOAK train/test splits for test subset=West



Significant difference between Same and Other?

Significant difference between Same and All?

— How can we get best prediction in the West?

— P-value from paired T-test (more powerful).

— Fold and subset are two different variables in data.

— Any number of folds possible (K=4 here for example).

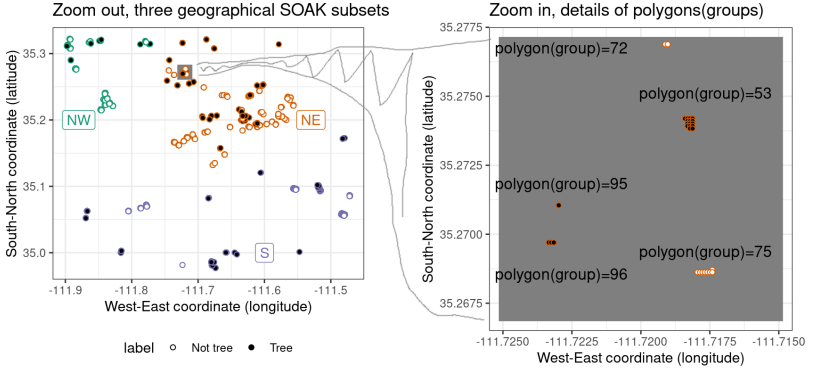
Previous (1 for loop) and proposed (3 for loops)

```
for test_fold from 1 to K: (previous standard/blocked CV)
  train_data = data[fold!=test_fold]
  test_data = data[fold==test_fold]
compute mean test error over folds
No P-values for comparing Same/Other/All training!

for test_subset in subsets:                (Proposed SOAK)
  for model in Same,Other,All:
    train_subsets = get_train_subsets(test_subset, model)
    for test_fold from 1 to K:
      train_data = data[fold!=test_fold, train_subsets]
      test_data = data[fold==test_fold, test_subset]
    compute mean test error over folds
  compute P-values, Same-Other and Same-All
return P-values for each test subset
```

Proposed SOAK algorithm new to ML frameworks

- Previous scikit-learn/mlr3: when splitting, rows in a **group** stay together, **strata** proportions preserved.
- For example (below, Shenkin *et al.*), satellite image segmentation, trees vs background (**strata**), rows=pixels, **grouped by polygon**, SOAK **subsets** are geographical regions (NW, NE, S).
- SOAK: good predictions on one test **subset**, after training on Same/Other/All subsets? (can use together with groups/strata)



Previous work and novelty of SOAK

- Free software machine learning frameworks: scikit-learn (Pedragosa 2011) and mlr3 (Bischl 2019) support group/strata column roles, **not subset**.
- Spatial/blocked CV similar to SOAK "Other" but **without distinct subset column, only one split per test block**. Wenger and Olden (2012), Roberts *et al.* (2017), Valavi *et al.* (2019), Ploton *et al.* (2020), Schratz *et al.* (2024).
- Same/Other comparison **without P-values** for out-of-distribution predictions (Madan *et al.*, 2020).
- Farahani *et al.* (2021) review domain adaptation, Other model is "unsupervised" because no labels available in the target domain (**missing comparison with Same**).
- **Proposed Same/Other/All K-fold cross-validation (SOAK, 2026) + mlr3resampling R package: new subset column distinct from previous fold/group/stratum columns, several train/test splits per test subset, P-values for Same-Other and Same-All comparisons**.

ML framework feature comparison

	R		python	
	mlr3	how?	scikit-learn	how?
Learners	multiple	benchmark_grid()	single	cross_validate()
Subsets(proposed)	yes	mlr3resampling	no	code it yourself
Parallelization	yes	mlr3batchmark	no	code it yourself

Support for **structured** cross-validation:

- scikit-learn StratifiedGroupKFold supports group and stratum roles, not subset.
- mlr3resampling supports three column roles: stratum, subset, and group, see vignette "Using subset with group and stratum." `https://mlr3resampling.netlify.app/doc/subset_group_stratum`

Support for **parallel** cross-validation:

- scikit-learn cross_validate() computes test error for a single algorithm (no special parallelization support).
- mlr3 supports parallelization over algorithm, data sets, and train/test splits via mlr3batchmark package (using all CPUs on your laptop, SLURM cluster, etc).

Introduction: how to evaluate machine learning algorithms?

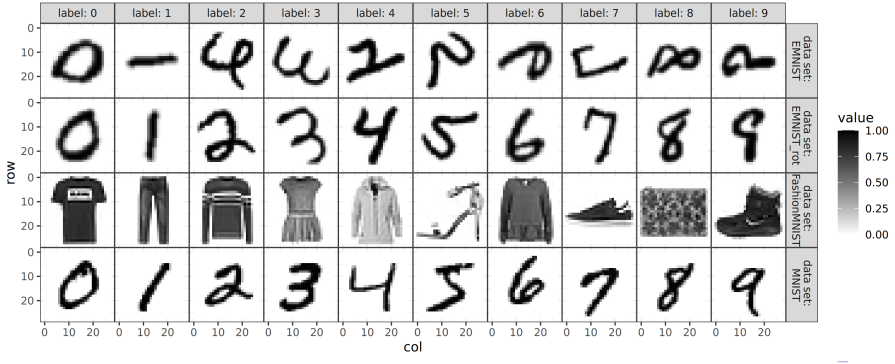
Proposed new algorithm, Same/Other/All K-fold cross-validation (SOAK)

Results on 20 data sets

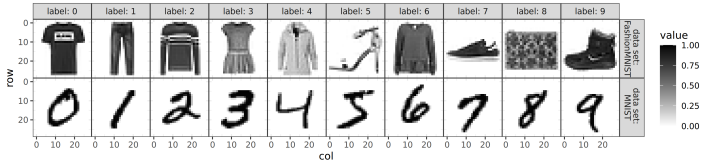
Conclusions

ImagePair data: train on MNIST and accurately predict on variants?

- Expect: $g($  $)$ and $h($  $)$ are not accurate.
- But train on MNIST, predict on EMNIST_rot (upright digits) should be possible, right? (Vote!) Let's use SOAK to find out!
- Create three new IPair data sets by combining MNIST with variants: EMNIST, EMNIST_rot, FashionMNIST.
- Each data set has two subsets: MNIST/variant.

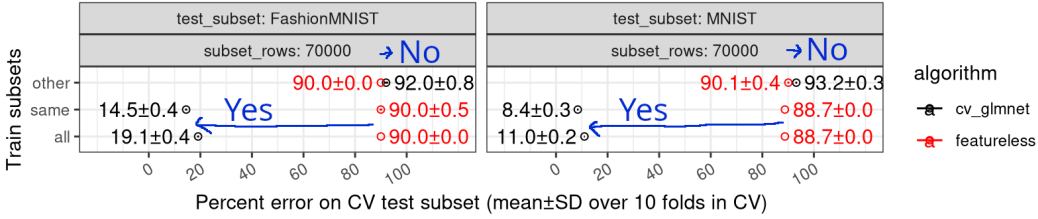


SOAK for IPair_Fashion data set (MNIST+FashionMNIST)



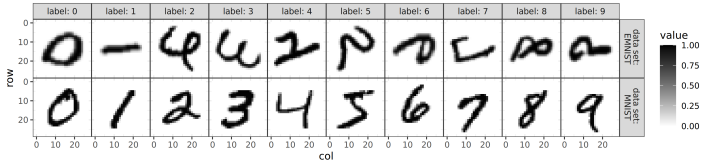
Data set: IPair_Fashion (MNIST+FashionMNIST)

Learning anything useful?

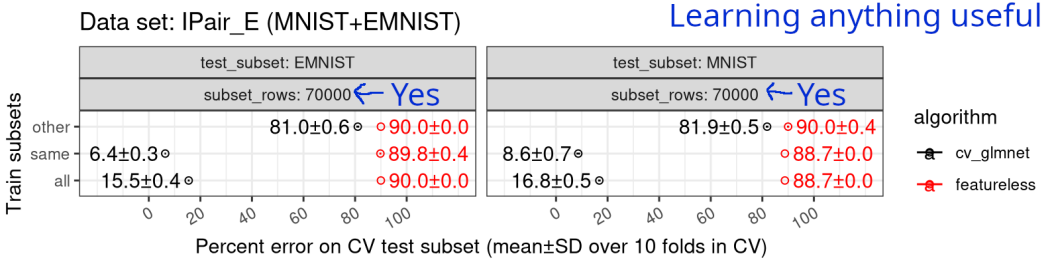


- **cv_glmnet** is L1-regularized linear model.
- **featureless** baseline always predicts most frequent class.
- **Other cv_glmnet** has greater test error than **featureless**.
- The patterns in the two subsets are too different for **Other cv_glmnet** to learn anything useful for prediction. (expected)

SOAK for IPair_E data set (MNIST+EMNIST)

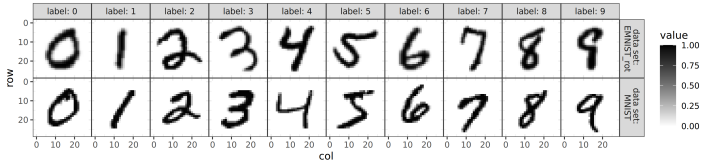


Learning anything useful?



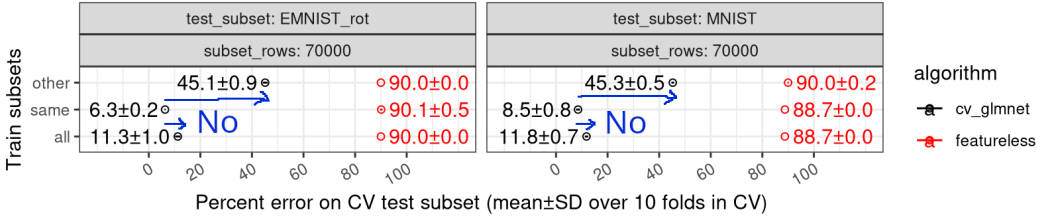
- **Other cv_glmnet** has significantly smaller test error than **featureless**, so linear model learns something that useful for predicting on the other subset.
- But since **Other** error rate is much larger than **Same**, clearly the pattern is very different between MNIST/EMNIST.

SOAK for IPair_E_rot data set (MNIST+EMNIST_rot)



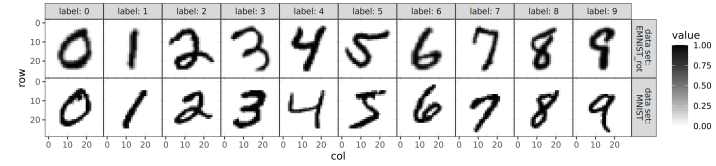
Data set: IPair_E_rot (MNIST+EMNIST_rot)

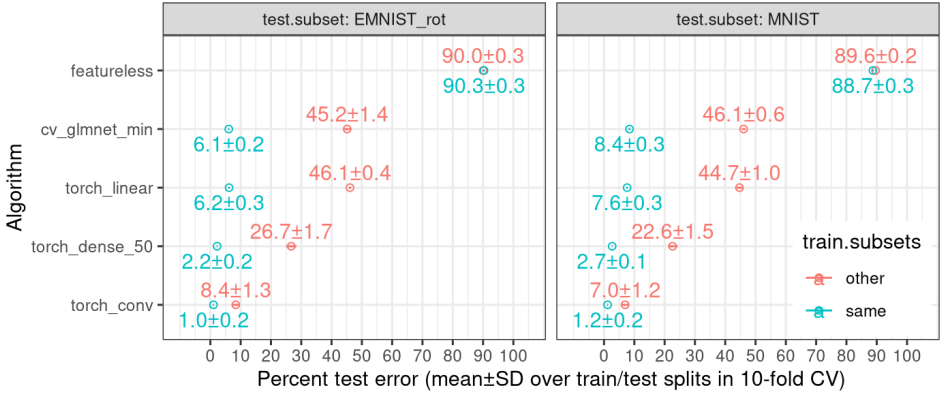
Other/All as accurate as Same?



- **Other cv_glmnet** has even smaller test error, indicating even more similarity between MNIST and EMNIST_rot data sets.
- But **Other/All** test error are still not as small as **Same**.
- Significant difference between patterns learnable/predictable by linear model in MNIST/EMNIST_rot. (surprising)

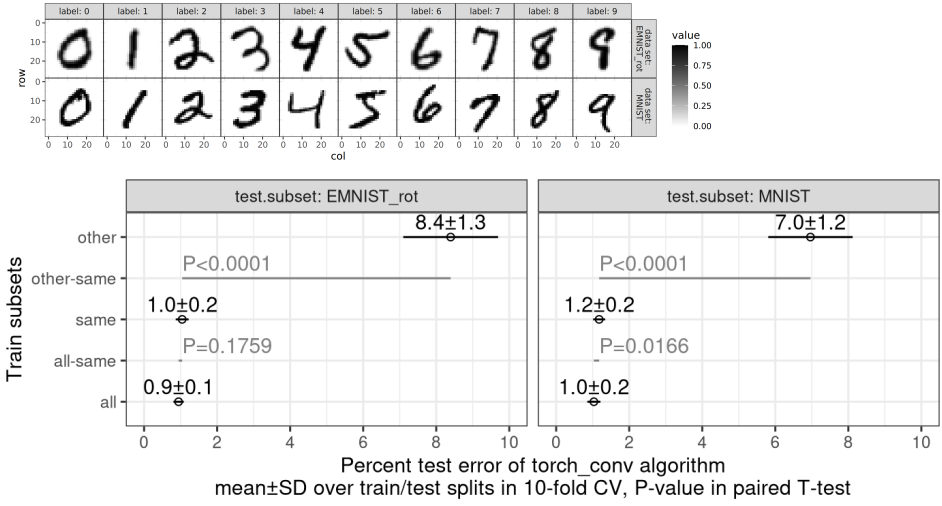
SOAK for IPair_E_rot data set (MNIST+EMNIST_rot), several learners





Neural networks (torch_dense_50, torch_conv) have smaller test error than linear models, but **Same** still has smaller test error than **Other**.

SOAK for IPair_E_rot data set (MNIST+EMNIST_rot), conv. net

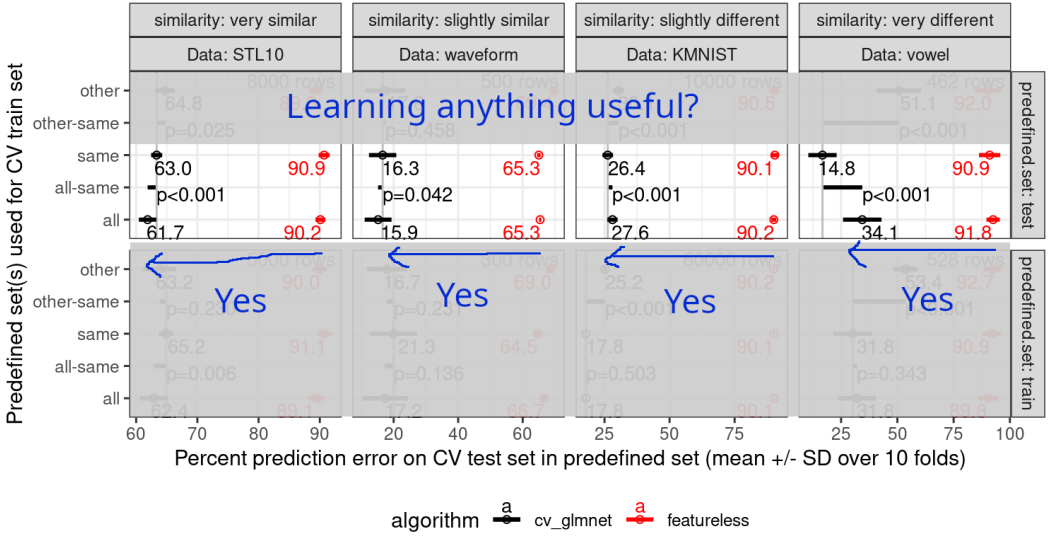


- For torch_conv (convolutional neural network), test error for **Other** is still larger than **Same** (can't train on one subset and predict optimally on the other).
- But test error for **All** is smaller than **Same** — can combine subsets when training, for more accurate predictions.

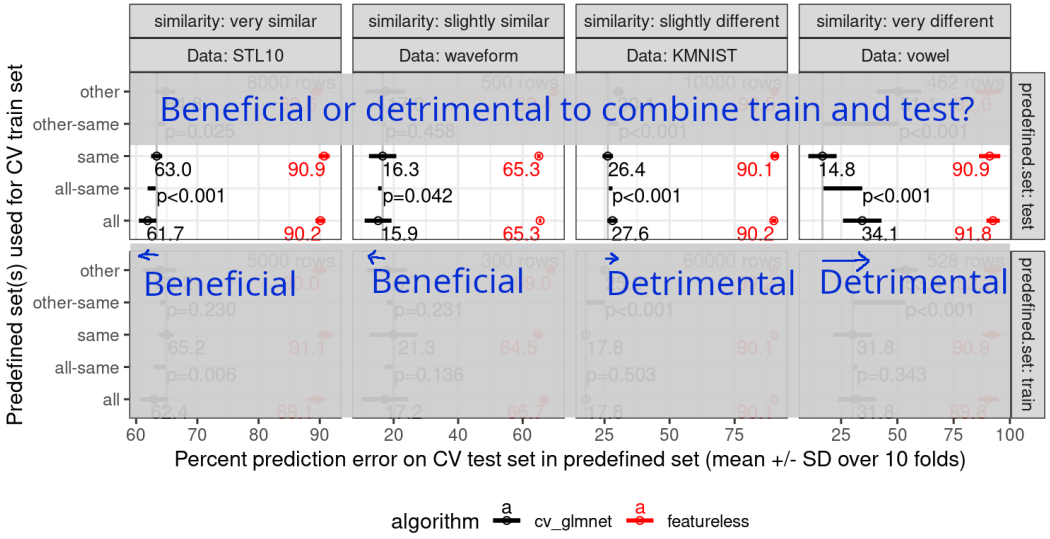
Benchmark data with pre-defined train/test subsets

- Machine learning researchers evaluate new algorithms using benchmark data sets, which sometimes have pre-defined train/test subsets.
- For example KMNIST is an image classification data set with
  60,000 images in a pre-defined train subset, and
  10,000 images in a pre-defined test subset.
- STL10 is another image classification data set with
  5000 images in a pre-defined train subset, and
  8000 images in a pre-defined test subset.
- Are the learnable/predictable patterns in the pre-defined train/test subsets similar? (expected if random sampling was used to create pre-defined subsets)
- Use pre-defined train/test subsets in SOAK, to see if patterns are learnable/predictable across pre-defined subsets.

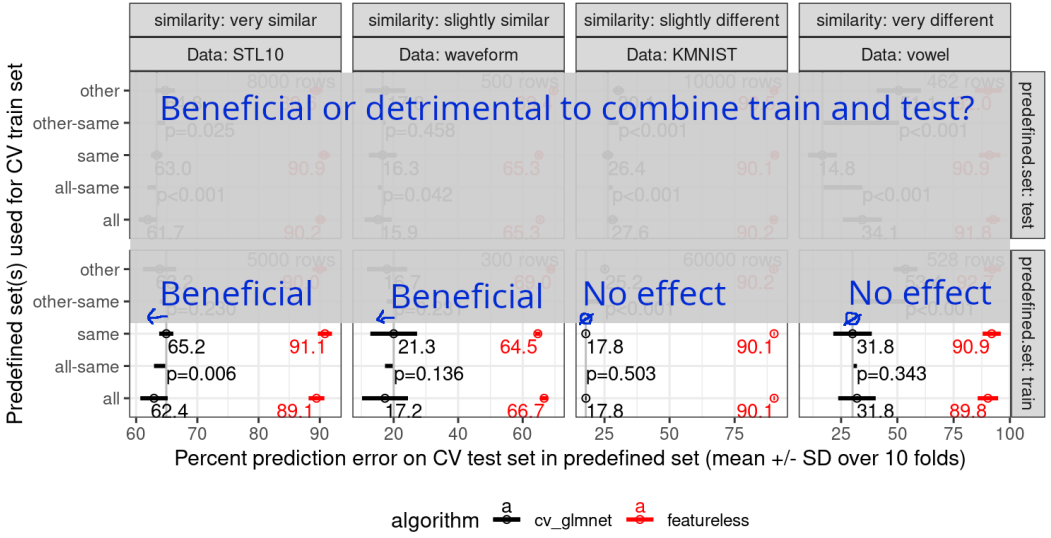
Benchmark data with pre-defined train/test subsets



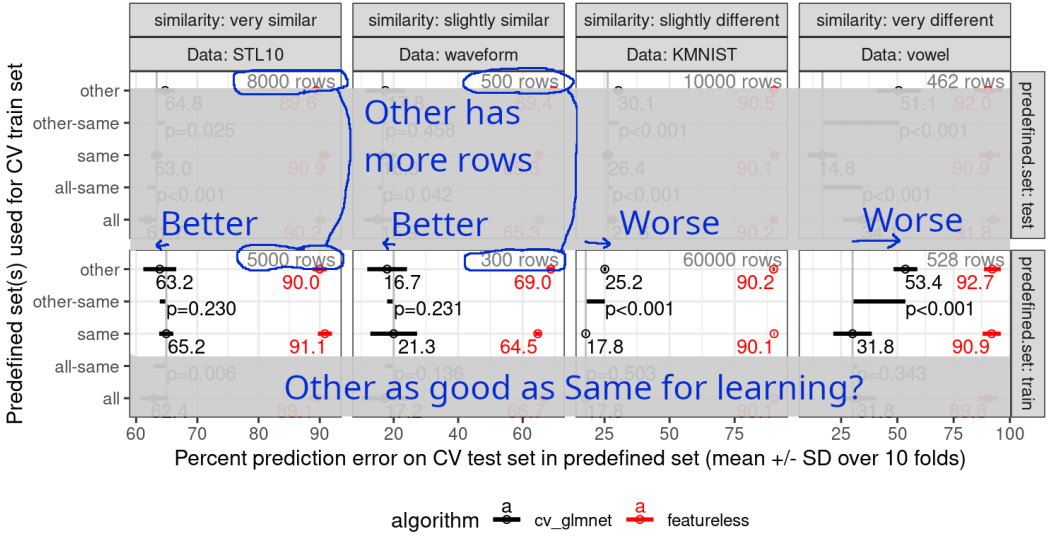
Benchmark data with pre-defined train/test subsets



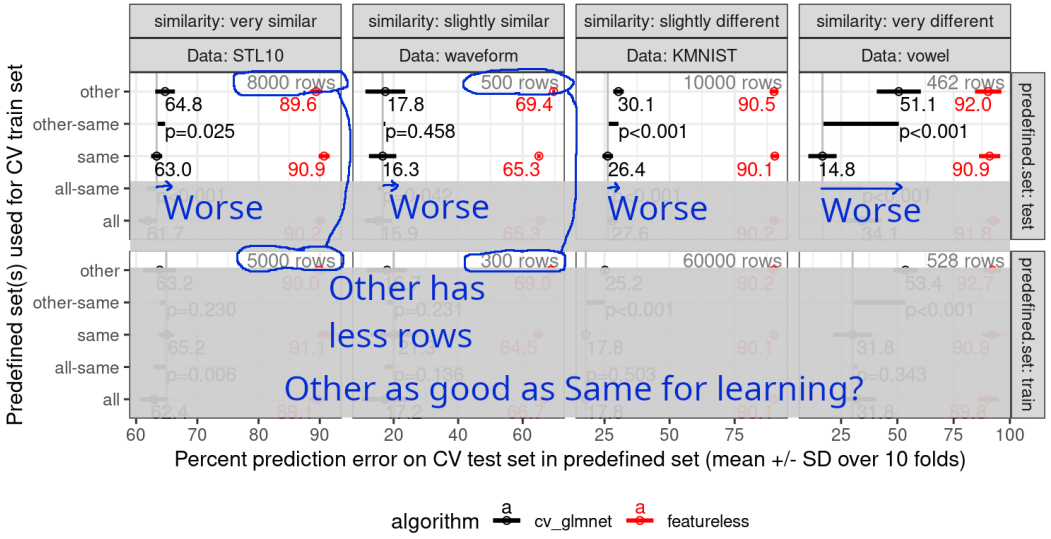
Benchmark data with pre-defined train/test subsets



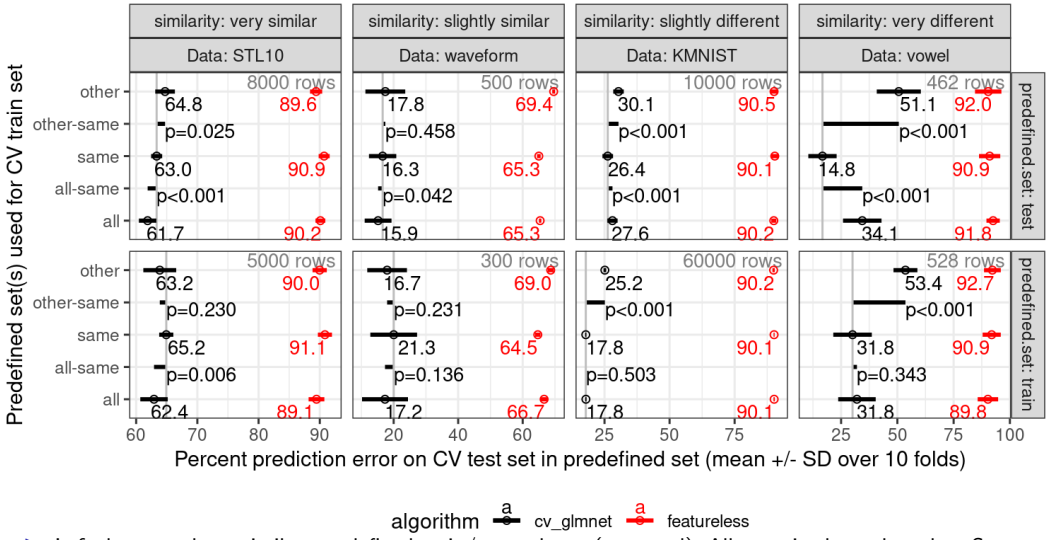
Benchmark data with pre-defined train/test subsets



Benchmark data with pre-defined train/test subsets



Benchmark data with pre-defined train/test subsets



- Left data sets have similar pre-defined train/test subsets (expected), All error is always less than Same, and Other error is either greater or less than Same, depending on subset sizes.
- Right data sets have different pre-defined train/test subsets (surprising), All error is always greater than or equal to Same, and Other error is always greater than Same.

20 classification data sets analyzed using SOAK

We considered MNIST and variants as subsets (**ImagePair**),
data from collaborations (**time/space** subsets),
and benchmark data with pre-defined **train/test** subsets.

	Type	Data	rows	features	classes	subsets	imb.
1	● ImagePair	IPair_E	140000	784	10	2	1.0
2	● ImagePair	IPair_E_rot	140000	784	10	2	1.0
3	● ImagePair	IPair_Fashion	140000	784	10	2	1.0
4	○ time/space	CanadaFiresA	4827	46	2	4	7.0
5	○ time/space	CanadaFiresD	1491	46	2	4	1.6
6	○ time/space	FishSonar_river	2815744	81	2	4	1.2
7	○ time/space	NSCH_autism	46010	364	2	2	1.5
8	○ time/space	aztrees3	5956	21	2	3	2.0
9	○ time/space	aztrees4	5956	21	2	4	4.9
10	● train/test	CIFAR10	60000	3072	10	2	5.0
11	● train/test	EMNIST	70000	784	10	2	6.0
12	● train/test	FashionMNIST	70000	784	10	2	6.0
13	● train/test	KMNIST	70000	784	10	2	6.0
14	● train/test	MNIST	70000	784	10	2	6.0
15	● train/test	QMNIST	120000	784	10	2	1.0
16	● train/test	STL10	13000	27648	10	2	1.6
17	● train/test	spam	4601	57	2	2	2.0
18	● train/test	vowel	990	10	11	2	1.1
19	● train/test	waveform	800	21	3	2	1.7
20	● train/test	zipUSPS	9298	256	10	2	3.6

All-Same comparison across 20 data sets, cv_glmnet algorithm

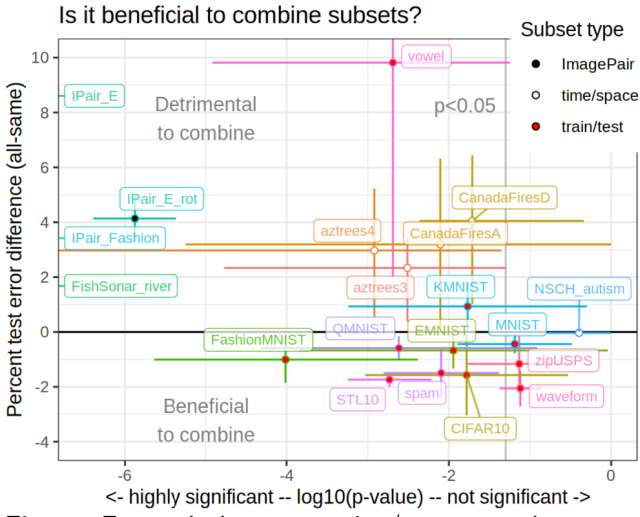


Figure: For each data set, point/segments show mean/range over test subsets.

Data	ErrorDiff	log10(P)
● IPair_E	8.2, 9.1	-11.2, -10.2
● IPair_E_rot	3.3, 5.0	-6.4, -5.4
● IPair_Fashion	2.6, 4.6	-12.4, -8.6
● vowel	2.0, 17.7	-4.9, -0.5
○ CanadaFiresD	1.0, 6.4	-2.4, -0.3
○ aztrees4	0.5, 5.2	-7.0, -1.4
○ FishSonar_river	0.5, 4.4	-14.4, -8.6
○ aztrees3	0.4, 3.4	-4.8, -1.3
● KMNIST	0.0, 1.8	-3.2, -0.3
○ CanadaFiresA	-0.0, 6.3	-5.2, 0.0
○ NSCH_autism	-0.1, -0.0	-0.8, -0.0
● MNIST	-0.8, -0.1	-1.9, -0.5
● QMNIST	-1.0, -0.2	-4.3, -0.9
● EMNIST	-1.3, -0.0	-3.8, -0.0
● FashionMNIST	-1.9, -0.2	-5.6, -2.4
● STL10	-2.0, -1.5	-3.2, -2.2
● zipUSPS	-2.2, -0.1	-1.8, -0.5
● spam	-2.4, -0.6	-2.8, -1.4
● waveform	-2.7, -1.4	-1.4, -0.9
● CIFAR10	-3.0, -0.1	-3.0, -0.5

Table: ranges over test subsets.

- Top: 10 data sets with incompatible subsets.
- Bottom: 10 data sets with compatible subsets.

Other-Same comparison across 20 data sets, cv_glmnet algorithm

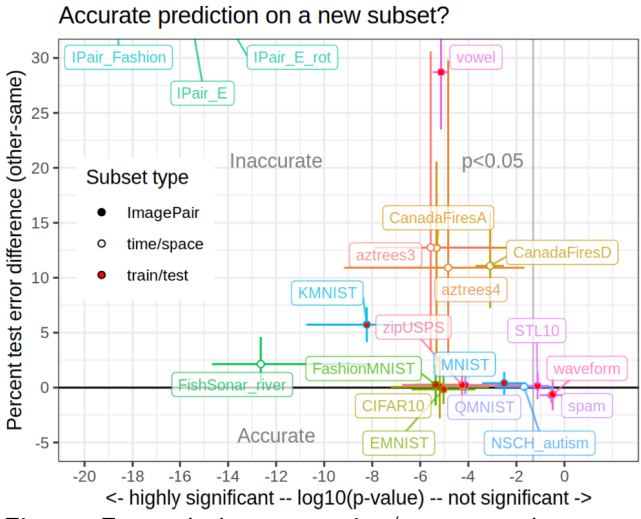


Figure: For each data set, point/segments show mean/range over test subsets.

Data	ErrorDiff	log10(P)
● IPair_Fashion	77.5, 84.7	-21.7, -17.8
● IPair_E	73.2, 74.7	-19.2, -18.3
● IPair_E_rot	36.7, 38.8	-15.3, -15.1
● vowel	23.5, 33.9	-5.5, -4.8
○ CanadaFiresD	7.2, 16.0	-3.7, -2.5
● KMNIST	4.1, 7.3	-10.7, -5.7
○ aztrees3	3.3, 30.6	-7.9, -2.9
○ aztrees4	3.0, 29.8	-9.2, -1.7
○ CanadaFiresA	2.4, 20.6	-6.7, -2.4
○ FishSonar_river	0.5, 4.6	-14.7, -8.1
○ NSCH_autism	-0.1, 0.2	-2.9, -0.5
● QMNIST	-0.6, 1.0	-4.9, -3.3
● MNIST	-0.6, 1.5	-3.4, -1.6
● STL10	-1.1, 1.4	-1.6, -0.6
● spam	-1.4, 0.1	-1.0, -0.0
● EMNIST	-1.5, 1.2	-6.4, -3.7
● FashionMNIST	-1.6, 2.2	-6.3, -4.5
● waveform	-2.0, 0.8	-0.6, -0.3
● zipUSPS	-2.1, 2.6	-6.8, -1.8
● CIFAR10	-2.8, 2.9	-7.2, -3.1

Table: ranges over test subsets.

- Top: 10 data sets with incompatible subsets.
- Bottom: 10 data sets with compatible subsets.

Introduction: how to evaluate machine learning algorithms?

Proposed new algorithm, Same/Other/All K-fold cross-validation (SOAK)

Results on 20 data sets

Conclusions

Summary of contributions

- Proposed Same/Other/All K-fold cross-validation (SOAK) estimates similarity of subsets (space, time, etc), for a particular learning algorithm.
- SOAK idea is new: P-values for Same-Other and Same-All comparisons, proposed **subset** column can be used with previous **group**/**stratum**.
- Free/open-source R package available in mlr3 framework `https://github.com/tdhock/mlr3resampling` (easy parallelization over algorithms, data sets, train/test splits)
- Ran SOAK in parallel over 20 classification data sets, 10 folds, 2–4 subsets, Same/Other/All (1200+ train/test splits).
- Train on MNIST, predict on FashionMNIST (impossible), predict on EMNIST (surprisingly difficult for linear model, even though both are digits).
- For linear model, most pre-defined train/test subsets in benchmark data were similar (STL10/waveform), two were not (KMNIST/vowel).
- We observed similarity between years in Autism data (linear model slightly benefits from combining years, predicting on new year works).
- In fires/trees/fish data, images/regions/rivers were too different for accurate learning and prediction using linear model.

Thanks! Please email me if you are interested to collaborate:
toby.dylan.hocking@usherbrooke.ca



Reproducible slide/figure source code:
`https://github.com/tdhock/cv-same-other-paper`

Learning algorithms we consider

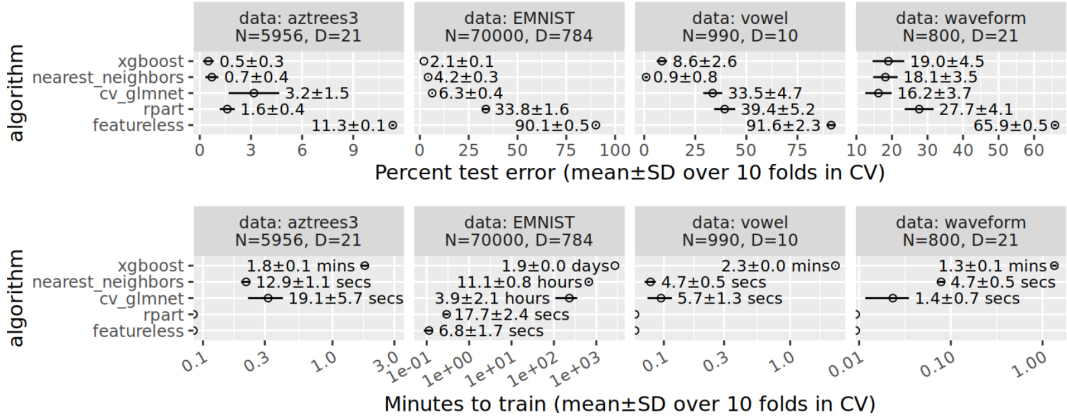
We used the following learning algorithms:

- cv_glmnet: L1-regularized linear model (feature selection). Friedman, *et al.* (2010).
- xgboost: Extreme gradient boosting (non-linear). Chen and Guestrin (2016).
- rpart: Recursive partitioning, decision tree (non-linear, feature selection). Therneau and Atkinson (2023).
- nearest_neighbors: classic non-linear algorithm, as implemented in kknn R package. Schliep and Hechenbichler (2016).
- featureless: un-informed baseline, ignores all inputs/features, and always predicts the most frequent label in train data. For example, Autism=No. Nomenclature from mlr3 R package, Lang, *et al.*, (2019).

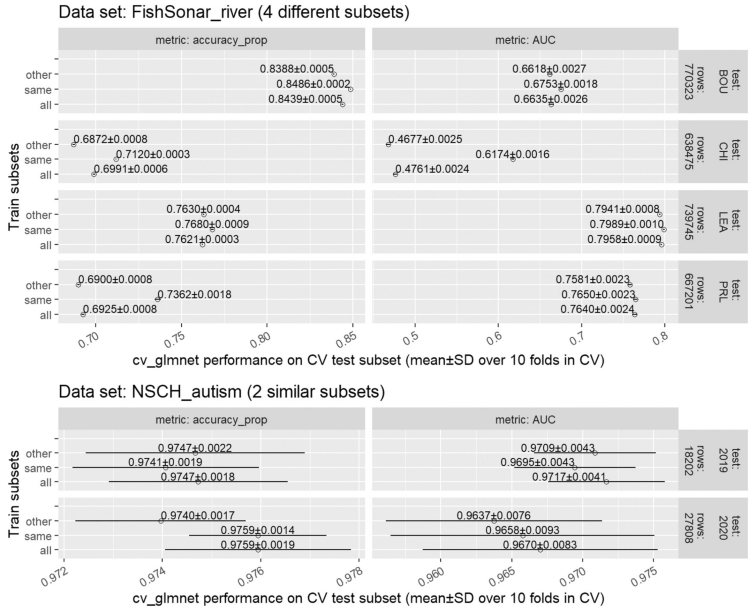
Each learning algorithm has different properties (non-linear, feature selection, etc). For details see Hastie, *et al.* (2009) textbook.

Learning algorithms have different accuracy and speed

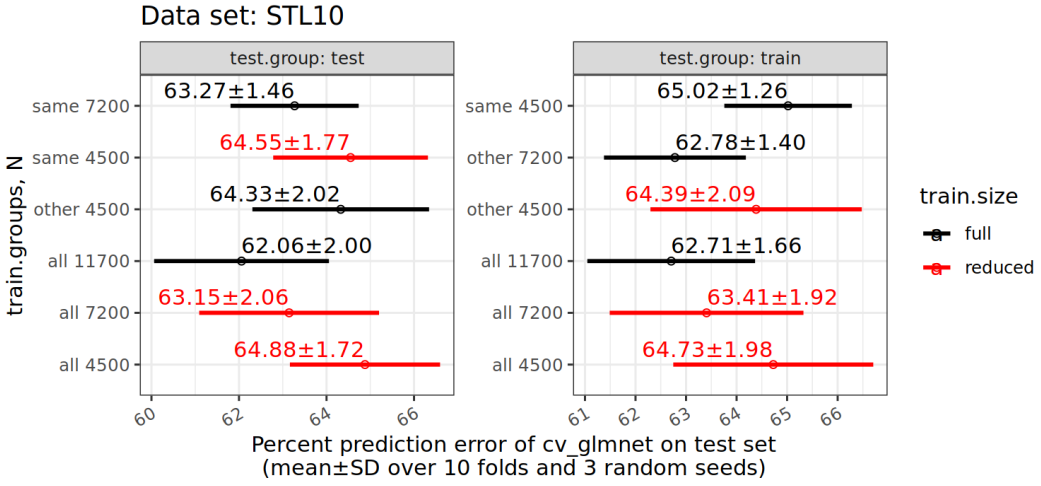
- Image segmentation/classification, and audio classification data sets.
- xgboost is slowest to train on all four data sets, most accurate in aztrees3, EMNIST.
- Other algorithms are faster and more accurate in vowel, waveform.
- Need to try a variety of algorithms for optimal prediction in any data.



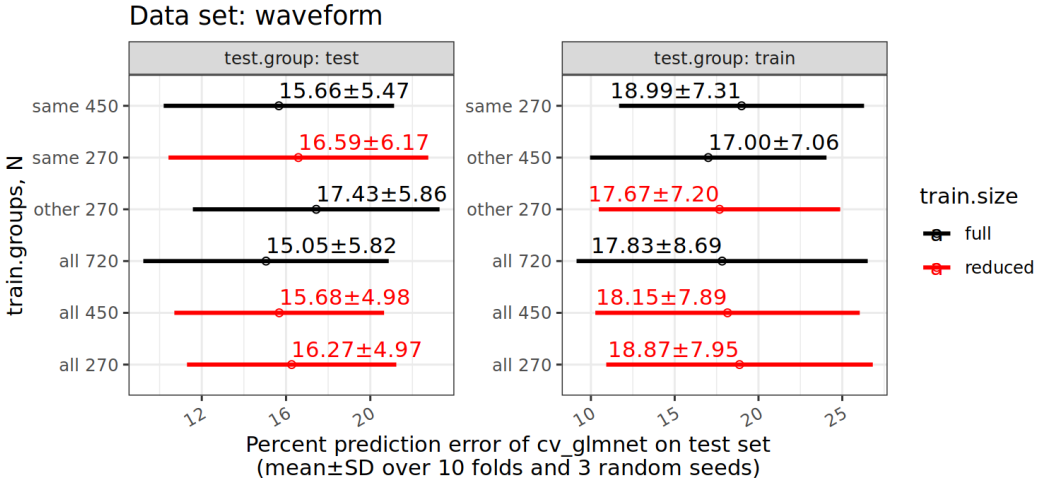
SOAK yields consistent conclusions using accuracy and AUC



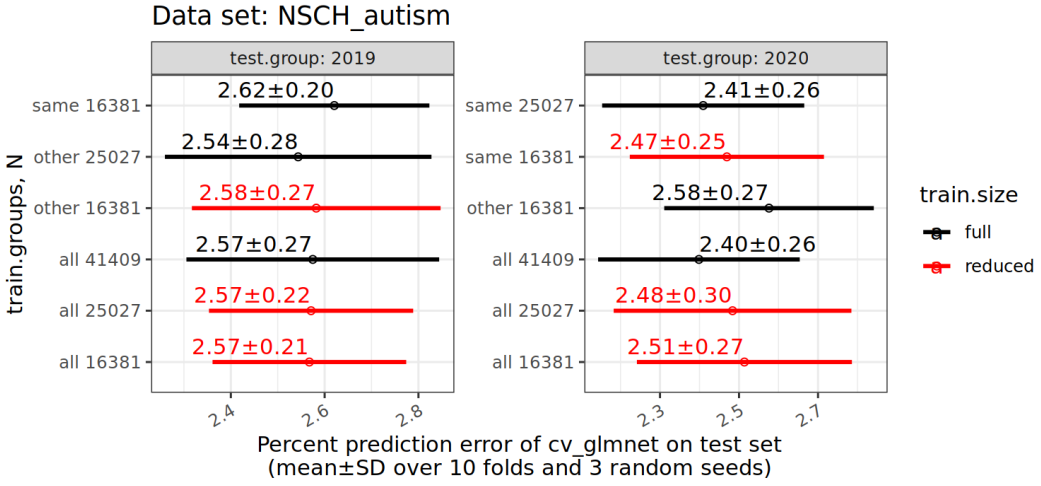
Large sample size effect

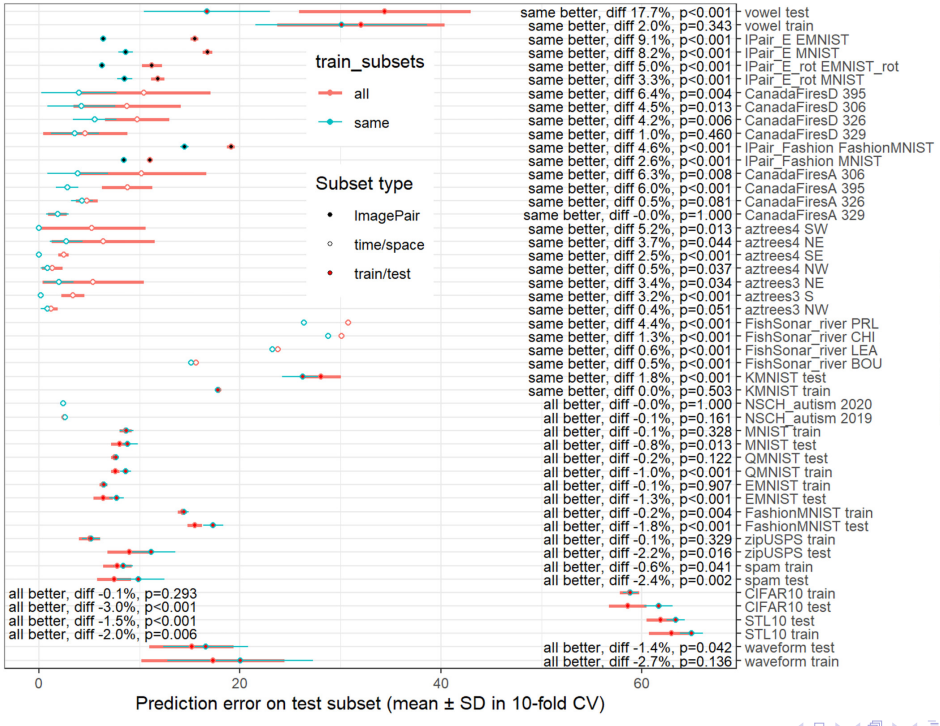


Moderate sample size effect



Small sample size effect



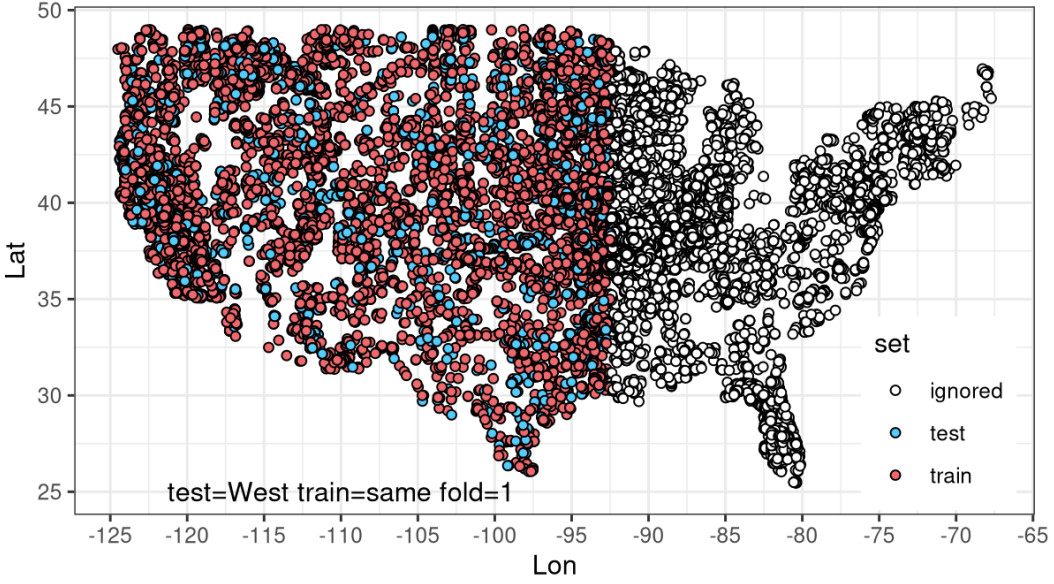

train_subsets
all
same
Subset type
ImagePair
time/space
train/test
all better, diff -0.1%, p=0.293
all better, diff -3.0%, p<0.001
all better, diff -1.5%, p<0.001
all better, diff -2.0%, p=0.006
same better, diff 17.7%, p<0.001 vowel test
same better, diff 2.0%, p=0.343 vowel train
same better, diff 9.1%, p<0.001 IPair_E EMNIST
same better, diff 8.2%, p<0.001 IPair_E MNIST
same better, diff 5.0%, p<0.001 IPair_E_rot EMNIST_rot
same better, diff 3.3%, p<0.001 IPair_E_rot MNIST
same better, diff 6.4%, p=0.004 CanadaFiresD 395
same better, diff 4.5%, p=0.013 CanadaFiresD 306
same better, diff 4.2%, p=0.006 CanadaFiresD 326
same better, diff 1.0%, p=0.460 CanadaFiresD 329
same better, diff 4.6%, p<0.001 IPair_Fashion FashionMNIST
same better, diff 2.6%, p<0.001 IPair_Fashion MNIST
same better, diff 6.3%, p=0.008 CanadaFiresA 306
same better, diff 6.0%, p<0.001 CanadaFiresA 395
same better, diff 0.5%, p=0.081 CanadaFiresA 326
same better, diff -0.0%, p=1.000 CanadaFiresA 329
same better, diff 5.2%, p=0.013 aztrees4 SW
same better, diff 3.7%, p=0.044 aztrees4 NE
same better, diff 2.5%, p<0.001 aztrees4 SE
same better, diff 0.5%, p=0.037 aztrees4 NW
same better, diff 3.4%, p=0.034 aztrees3 NE
same better, diff 3.2%, p<0.001 aztrees3 S
same better, diff 0.4%, p=0.051 aztrees3 NW
same better, diff 4.4%, p<0.001 FishSonar_river PRL
same better, diff 1.3%, p<0.001 FishSonar_river CHI
same better, diff 0.6%, p<0.001 FishSonar_river LEA
same better, diff 0.5%, p<0.001 FishSonar_river BOU
same better, diff 1.8%, p<0.001 KMNIST test
same better, diff 0.0%, p=0.503 KMNIST train
all better, diff -0.0%, p=1.000 NSCH_autism 2020
all better, diff -0.1%, p=0.161 NSCH_autism 2019
all better, diff -0.1%, p=0.328 MNIST train
all better, diff -0.8%, p=0.013 MNIST test
all better, diff -0.2%, p=0.122 QMNIST test
all better, diff -1.0%, p<0.001 QMNIST train
all better, diff -0.1%, p=0.907 EMNIST train
all better, diff -1.3%, p<0.001 EMNIST test
all better, diff -0.2%, p=0.004 FashionMNIST train
all better, diff -1.8%, p<0.001 FashionMNIST test
all better, diff -0.1%, p=0.329 zipUSPS train
all better, diff -2.2%, p=0.016 zipUSPS test
all better, diff -0.6%, p=0.041 spam train
all better, diff -2.4%, p=0.002 spam test
CIFAR10 train
CIFAR10 test
STL10 test
STL10 train
all better, diff -1.4%, p=0.042 waveform test
all better, diff -2.7%, p=0.136 waveform train
Data and test subset
0
20
40
60
Prediction error on test subset (mean ± SD in 10-fold CV)

Stratified fold assignment, 10-fold CV

```
> folds <- function(by){set.seed(1);
+ data.table(label=rep(0:1,c(100,900)))[
+ , fold := sample(rep(1:10,l=.N)), by
+ ][, table(label, fold)]}
> folds()
     fold
label  1  2  3  4  5  6  7  8  9 10
    0 13 13  4 10 13  7 12  8  9 11
    1 87 87 96 90 87 93 88 92 91 89
> folds("label")
     fold
label  1  2  3  4  5  6  7  8  9 10
    0 10 10 10 10 10 10 10 10 10 10
    1 90 90 90 90 90 90 90 90 90 90
```

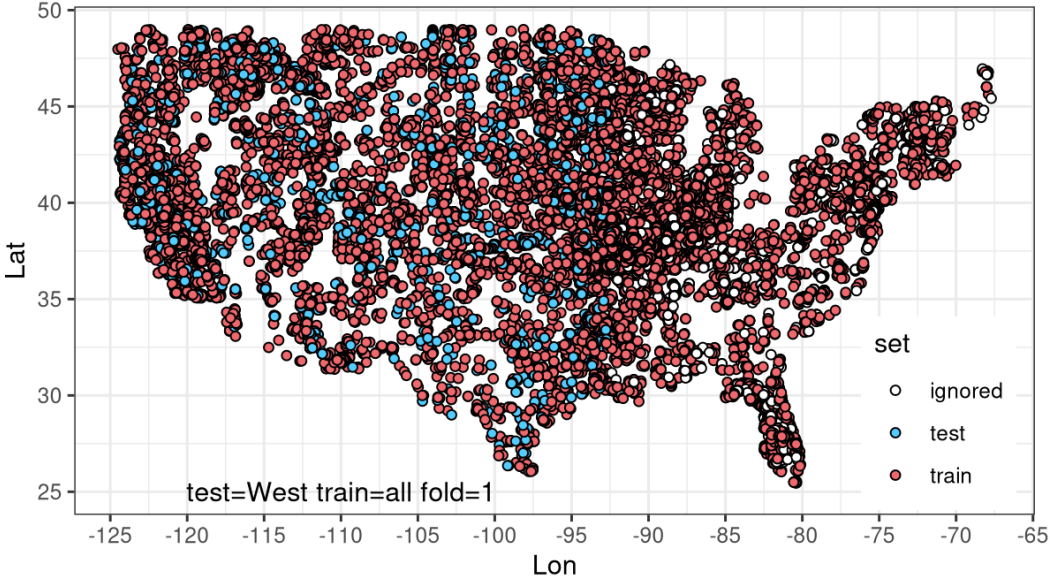
With stratification by label, we get the same label counts in each fold.

Proposed SOAK algorithm (geographical data example)



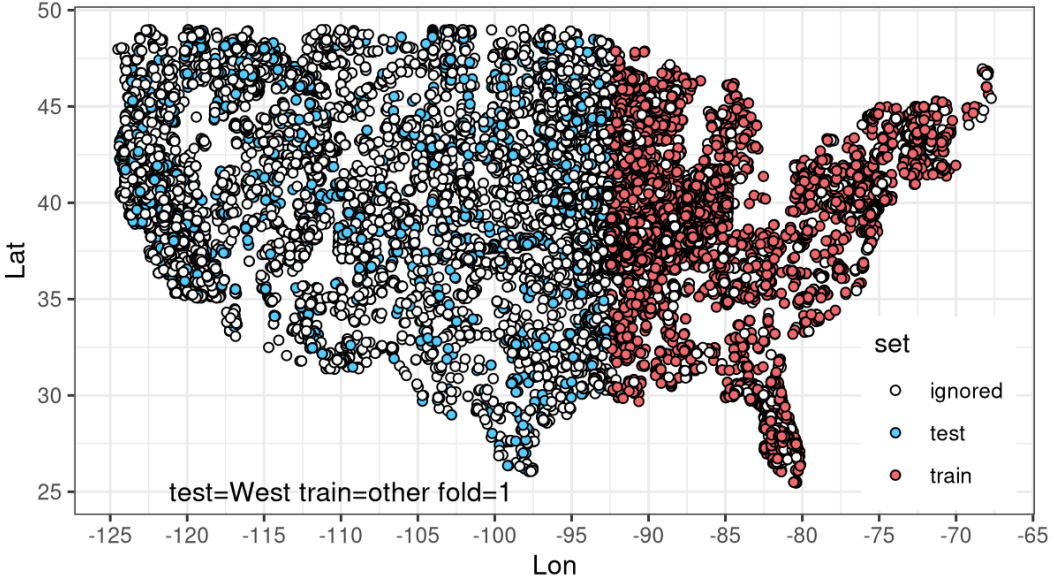
- Say we want good prediction in western USA.
- One of the K test sets is: subset=West and fold=1.

Proposed SOAK algorithm (geographical data example)



- Train on All (West+East, fold$\neq$1) would be less accurate than Same if subsets are different, and more accurate if subsets are similar (more relevant training data).

Proposed SOAK algorithm (geographical data example)



- Train on Other (East, fold$\neq$1) would be less accurate than Same if subsets are different.